LESSON PLAN

A/C STRUCTURE (11)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	02
Total No. of Lesson Plans	17	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Steps in aircraft design, manufacturing process and aircraft design requirements
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Steps in aircraft design, manufacturing process and aircraft design requirements
5. Specific Objectives	11.1.1.1 Trainee will be able to explain aircraft structure, various structural elements and their details.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of design steps and manufacturing phases; Safety: Maintain classroom discipline; Supervision: Instructor monitors engagement
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the primary structural elements of an aircraft?
(b) Can you list the initial steps in the aircraft design process?
(c) What are the basic requirements for aircraft design?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Conceptualization and preliminary design • Detailed design and analysis • Prototyping and testing phases • Manufacturing process overview • Quality control and final assembly	L & Ds	PPT, IB	15 min
(b)	• Aircraft design requirements and constraints • Material selection and structural elements • Integration of systems and subsystems • Regulatory compliance and standards • Cost estimation and schedule management	L & Ds	PPT, PDF	20 min
(c)	• Final inspection and certification steps • Maintenance planning and documentation • Lifecycle management and upgrades • Environmental impact considerations • Future trends in aircraft design	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the relationship between design requirements and manufacturing processes.
(b) How do structural elements influence the overall aircraft design?

12. Home Assignment and Follow Up	Prepare a brief note on how the steps in aircraft design lead to airworthiness certification requirements covered in LP 03.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Airworthiness and certification requirements of aircraft	Refer to the master syllabus reference list	LP 03

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min